

Markov Decision Processes

From finite to infinite horizon, on one three-state example: cycles, backward induction, discounting, value and policy iteration

SETTING

Four assumptions for the whole lecture

Finite states and actions

V is a vector, P an array; the max over actions exists; finitely many policies.

The state is observed

The agent sees s , and s holds everything that matters for the future (Markov).

The model is known

$P(s' | s, a)$ and $R(s, a, s')$ are given: this is planning, not learning.

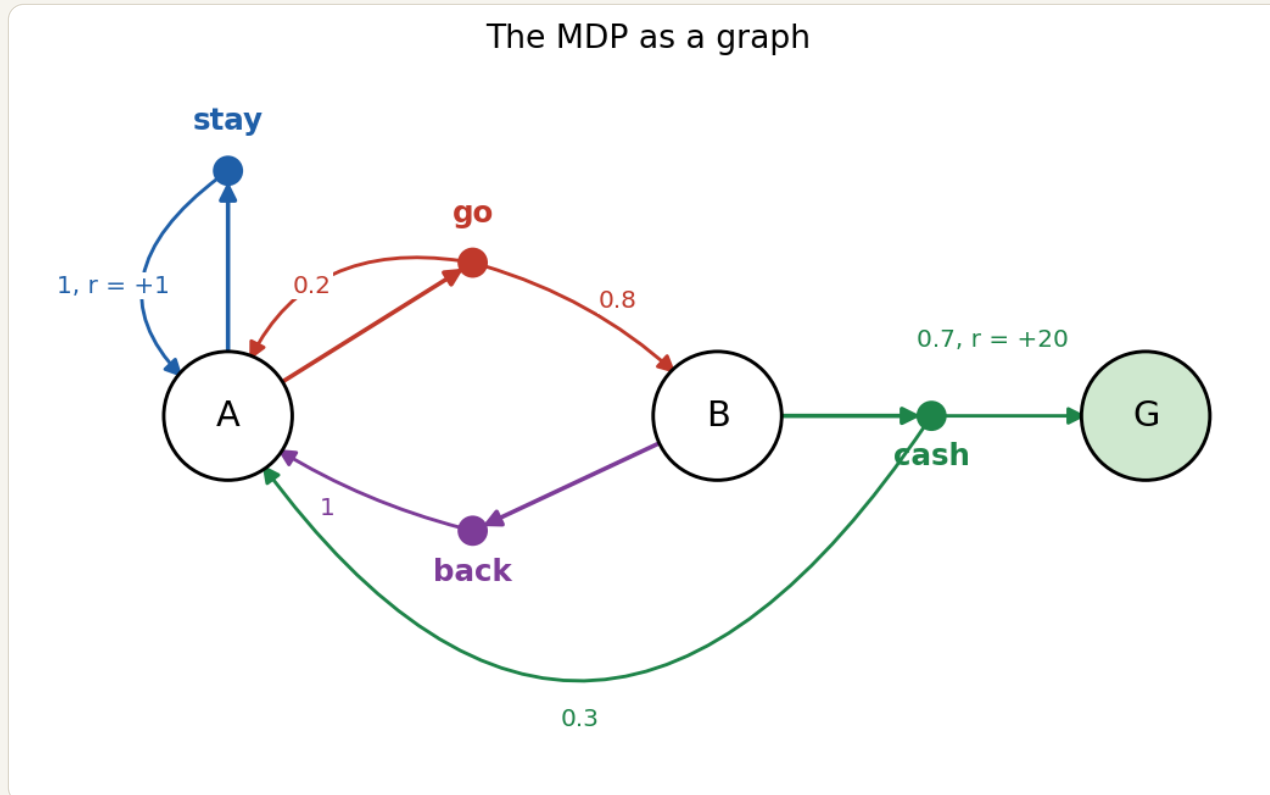
The goal is expected return

Maximize $E[r_1 + \gamma r_2 + \gamma^2 r_3 + \dots]$; something must keep this sum finite.

Later in the course we remove them one at a time: unknown model $\rightarrow$ learning; large states $\rightarrow$ function approximation; hidden state $\rightarrow$ POMDPs.

THE EXAMPLE

Three states, four actions, and cycles



stay in A: +1, stay in A

go from A: reach B with 0.8, slip back with 0.2

cash in B: +20 and end with 0.7; fail back to A with 0.3

back from B to A

G is terminal: absorbing, reward 0.

An episode samples $a \sim \pi(\cdot|s)$, then $s' \sim P(\cdot|s,a)$, until G. Its return

A policy as a conditional distribution

reward for ending up in s' taking action a from the state s :

$$R(s, a, s') = E[r|s, a, s']. \quad (1)$$

Note that

$$p(a, s', r|s) = \pi(a|s)p(s'|s, a)p(r|s, a, s'). \quad (2)$$

Using the law of iterated expectation twice, one has

$$\begin{aligned} E[r|s] &= \sum_r rp(r|s) = \sum_r r \sum_{a, s'} p(a, s', r|s) \\ &= \sum_a \pi(a|s) \sum_{s'} p(s'|s, a) \sum_r rp(r|s, a, s') \\ &= \sum_a \pi(a|s) \sum_{s'} p(s'|s, a) R(s, a, s') = E \left[E[E[r|s, a, s']|s'] \middle| a \right]. \end{aligned} \quad (3)$$

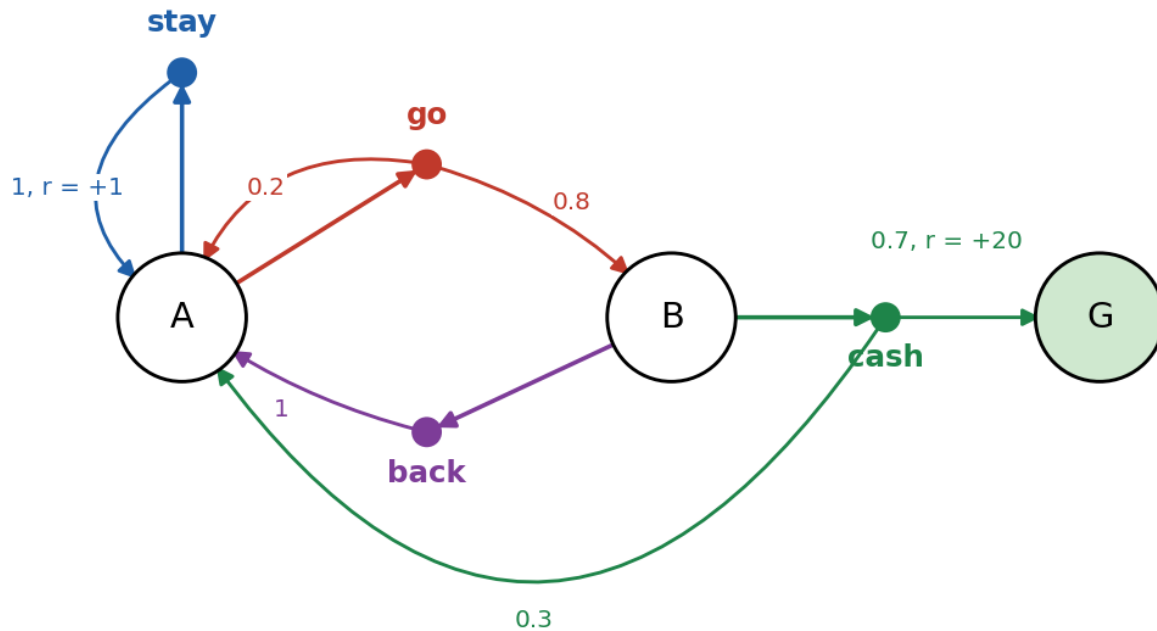
Observe that

$$E[r|s] \leq \max_a \sum_{s'} p(s'|s, a) R(s, a, s'). \quad (4)$$

POLICIES AND RETURNS

Trajectory, History, and Markov Property

The MDP as a graph



Stochastic: the general case

$$\pi(a \mid s)$$

Example: uniform, $\pi(\text{stay} \mid A) = \pi(\text{go} \mid A) = \frac{1}{2}$.

Deterministic: a special case

$$a = \pi(s)$$

All mass on one action. Example: go in A, cash in B.

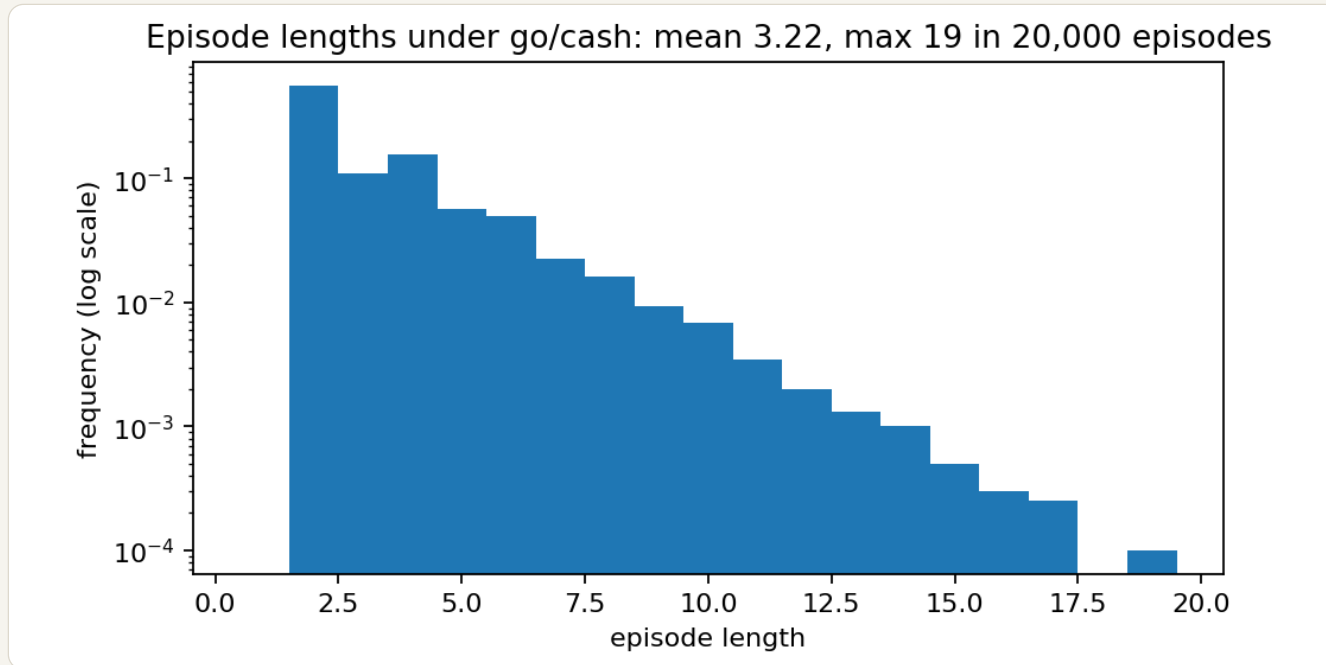
$$p(s_{t+1}, r_{t+1} \mid h_t, a_t) = p(s_{t+1}, r_{t+1} \mid s_t, a_t).$$

$$s_0, a_0, r_1, s_1, a_1, r_2, s_2, \dots, \quad a_t = \pi(s_t), \quad s_{t+1} \sim p(\cdot \mid s_t, a_t), \quad r_{t+1} \sim p(\cdot \mid s_t, a_t, s_{t+1})$$

$$h_t = (s_0, a_0, r_1, \dots, r_t, s_t)$$

EPISODES

A terminal state ends episodes, but not at a fixed time



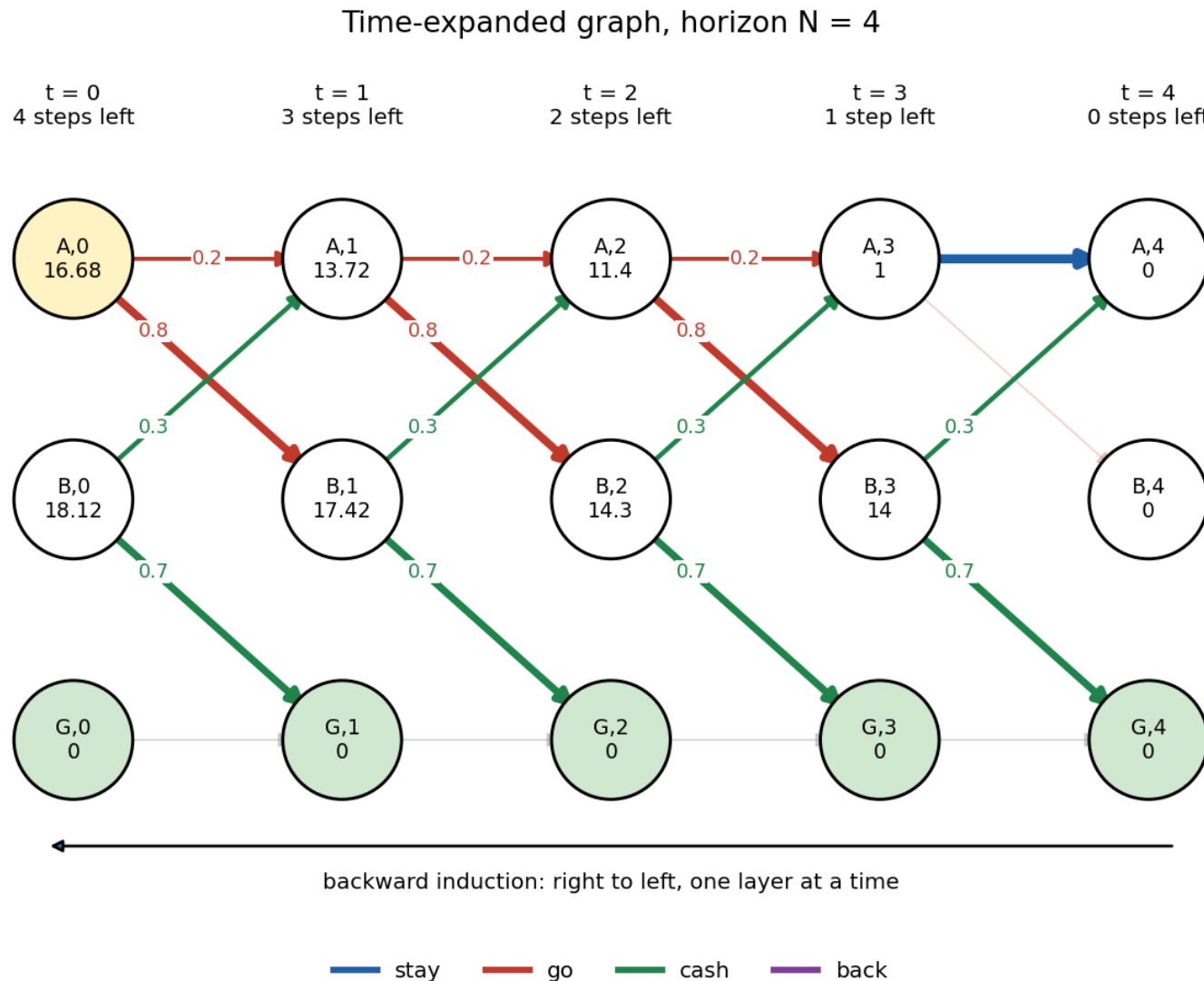
Under **go/cash** every episode ends with probability 1, but there is **no bound N** on its length.

Under **stay** forever it **never** ends: with $\gamma = 1$ the return after 10, 100, 1000 steps is 10, 100, 1000.

Such a policy is **improper**. Something must keep the sum finite.

FINITE HORIZON $N = 4$

Unrolled in time, the graph has no cycles. Backward induction



Every arrow goes from t to $t + 1$, so the graph is a DAG, even though the MDP has cycles.

A value is a probability-weighted average of the next column: $0.8 \times 14 + 0.2 \times 1 = 11.4$.

With 1 step left, A plays **stay**; otherwise **go**.

$$V_k(s) = \max_a \sum_{s'} p(s'|s, a) [R(s, a, s') + V_{k-1}(s')]$$

$$V_0(s) = 0$$

FINITE HORIZON, $\gamma = 1$

The optimal action depends on the time left

steps left k	1	2	3	4	5	6	7	8	9	10	11	12
$\pi_k(A)$	stay	go	go	go	go	stay	stay	stay	stay	stay	stay	stay
$\pi_k(B)$	cash	cash	cash	cash	cash	cash	cash	cash	back	back	back	back
$U_k(A)$	1.0	11.4	13.7	16.7	17.8	18.8	19.8	20.8	21.8	22.8	23.8	24.8

Far from the deadline, collect +1 by staying; cash in later.
Near it, rush for the +20.

$U_k(A)$ grows by about 1 per step: with no deadline and $\gamma = 1$ the value would be infinite.

INFINITE HORIZON

Drop the deadline, but keep the sum finite

No natural deadline

Control, balancing, trading: a fixed N invents end-of-time effects.

Stationary policy

One table $\pi(s)$, not one per time step.

Cleaner math

A fixed-point equation instead of a recursion.

What keeps the return finite	Horizon	Optimal policy
finite horizon N	fixed	depends on time left
terminal state, every policy proper	random, unbounded	stationary
discount $\gamma < 1$	random, mean $1/(1 - \gamma)$	stationary

$$G_t = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \dots$$

$$V_k(s) = \max_a \sum_{s'} p(s'|s, a) [R(s, a, s') + V_{k-1}(s')]$$

$$V^\pi(s) = \sum_{s'} p(s'|s, \pi(s)) [R(s, \pi(s), s') + \gamma V^\pi(s')]$$

$$V^\pi(s) = Q^\pi(s, \pi(s))$$

$$Q^\pi(s, a) = \sum_{s'} p(s'|s, a) [R(s, a, s') + \gamma V^\pi(s')]$$

DISCOUNTING

Discounting is a random deadline

After every step the process ends with probability $1 - \gamma$. The reward of step t is collected only if we survived t coin flips: probability γ^t , exactly its discount weight.

16.115

discounted return, $\gamma = 0.9$

16.116

undiscounted, with sudden death

10

mean lifetime $1/(1 - \gamma)$

$$E[T] = 1/(1 - \gamma)$$

BELLMAN EQUATION FOR A POLICY

With cycles, values are solved together

$$V^\pi(s) = \sum_{s'} p(s'|s, \pi(s)) [R(s, \pi(s), s') + \gamma V^\pi(s')]$$

$$R^\pi(s) = E[r|s] = \sum_{s'} p(s'|s, \pi(s)) R(s, \pi(s), s'), \quad P_{ss'}^\pi = p(s'|s, \pi(s)).$$

Then the Bellman expectation equation is the linear system

$$V^\pi = R^\pi + \gamma P^\pi V^\pi \implies V^\pi = (I - \gamma P^\pi)^{-1} R^\pi.$$

Optimal policy and the Bellman optimality equation

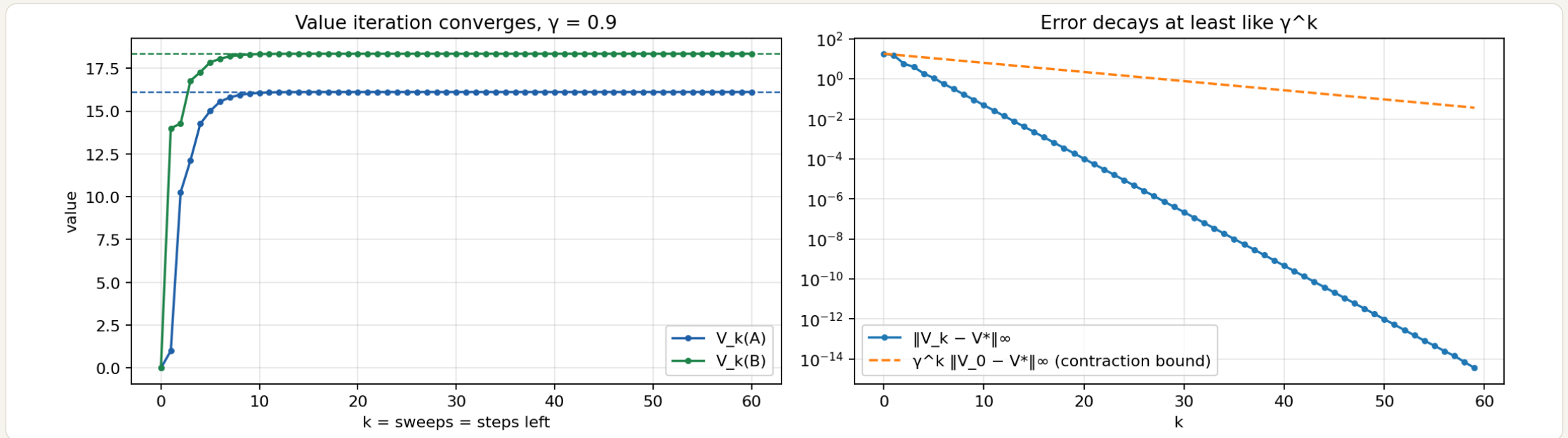
A policy π^* is *optimal* if $V^{\pi^*}(s) \geq V^\pi(s)$ for every state s and every policy π . The optimal values $V^* = V^{\pi^*}$ and $Q^* = Q^{\pi^*}$ satisfy the *Bellman optimality equation*

$$V^*(s) = \max_a \sum_{s'} p(s'|s, a) [R(s, a, s') + \gamma V^*(s')] = \max_a Q^*(s, a). \quad (16)$$

CONTRACTION

The Bellman update shrinks errors by γ every sweep

$$\|BV - BV'\|_{\infty} \leq \gamma \|V - V'\|_{\infty}$$



So V^* exists and is unique, value iteration converges from any start, and $\|V_k - V^*\| \leq \gamma^k \|V_0 - V^*\|$.

BELLMAN EQUATION FOR A POLICY

Policy Iteration Algorithm

1. *Evaluation*: solve $V^\pi = (I - \gamma P^\pi)^{-1} R^\pi$.
2. *Improvement*: $\pi'(s) = \arg \max_a Q^\pi(s, a)$.
3. If $\pi' = \pi$, stop; otherwise set $\pi \leftarrow \pi'$ and repeat.

Optimal policy and values, $\gamma = 0.9$

